



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (25/05/2026 – 31/05/2026)

WEEK 2

PROJECT NAME

**RADIO PROPAGATION ANALYSIS AND
PATH PREDICTION**

PROJECT DESCRIPTION :

Project focuses on simulating antenna beam propagation and predicting communication coverage.

GROUP MEMBER :

**GARIMA CHOUDHARY, MANASH PRASAR,
ARYAN KUMAR**

GROUP ID :

2026G265

GROUP GUIDE :

Dr. ABDUL JHUMMARWALA

GROUP COORDINATOR :

GARIMA CHOUDHARY

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ABHILASHA SAKSENA

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COLLEGE NAME :

GATI SHAKTI VISHWAVIDYALAYA

25/05/2026	• Studied 3D radiation-pattern concepts and explored methods for radiation-lobe visualization. (Garima choudhary) • Explored terrain datasets (SRTM/ASTER) and their role in propagation analysis. (Aryan kumar) • Studied GNU Octave radiation-pattern generation workflow for antenna arrays. (Manash prasar)
26/05/2026	• Explored Cesium Sandcastle environment and basic 3D visualization examples. (Garima choudhary) • Studied basic path-prediction and propagation-analysis concepts. (Aryan kumar) • Reviewed non-uniform antenna-array radiation patterns and gain behavior. (Manash prasar)
27/05/2026	• Generated a 3D radiation pattern and analyzed its lobe structure. (Garima choudhary) • Explored RF propagation tools and coverage-analysis examples. (Aryan kumar) • Studied relationship between gain, beamwidth, and propagation distance. (Manash prasar)
28/05/2026	Holiday (Bakrid)
29/05/2026	• Worked on understanding conversion of 3D radiation patterns into visualization-friendly formats. (Garima choudhary) • Studied terrain influence on signal propagation using available datasets. (Aryan kumar) • Simulated Yagi-Uda antenna in 4NEC2 and observed radiation-pattern characteristics. (Manash prasar)
30/05/2026	• Extracted and analyzed a 2D slice from the generated 3D radiation pattern for further study. (Garima choudhary) • Explored communication-coverage and path-prediction examples. (Aryan kumar) • Compared GNU Octave radiation patterns with 4NEC2 simulation outputs. (Manash prasar)
31/05/2026	Holiday (Sunday)

Next Week Plan	We plan to visualize antenna radiation patterns in Cesium Sandcastle and study the conversion of 3D radiation data into usable visualization formats. We will also continue terrain-data analysis and antenna-simulation studies for radio propagation and path prediction.
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REFERENCE:

<https://sandcastle.cesium.com/>

<https://cesium.com/learn/cesiumjs-learn/>

<https://www.qsl.net/4nec2/>

<https://www.antenna-theory.com/>

<https://www.qsl.net/kd2bd/splat.html/>

<https://www.earthdata.nasa.gov/data/instruments/aster>

<https://octave.org/doc/interpreter/>

SCREEN SHOTS:

